

the prime directive among investment managers is for them to keep their jobs. In order to do this, one should never be wrong on one's own, which creates herding among professional investment managers.

Charles MacKay wrote in his classic book of 1841, *Extraordinary Popular Delusions and the Madness of Crowds*, "Men, it has been well said, think in herds; it will be seen they go mad in herds, while they only recover their senses slowly, and one by one." Herding has a strong physiological, as well as psychological, basis. It is associated with the release of oxytocin and positive feelings of trust and security. Isolation from herds, on the other hand, leads to stimulation of the amygdala neurons, which can trigger a fight-or-flight response and overwhelm the analytical brain.¹

Herding is primordial. It manifests itself when an animal stays with the crowd in order to reduce its risk of attack. Herding is deeply ingrained in our brain chemistry and DNA.

There is also evidence that market activity can itself stimulate changes in physiology and create additional behavioral changes. Kandasamy et al. (2014) showed that investors experience a sustained increase in the stress hormone cortisol when market volatility increases, which causes them to become more risk-averse. Physiology-induced shifts in risk preferences may be an underappreciated cause of market instability. It may help explain why many individual investors tend to sell in herds at or near market bottoms. It may also give us a better understanding of the basis for feedback-related behavior. We will see in later chapters how absolute and dual momentum can help by removing falling assets from our portfolios before our stress levels become high enough to cause us to behave in ways that are injurious to our financial well-being.

There are several other theories advanced as to why investors follow positive-feedback strategies that lead to herding behavior. Barberis, Shleifer, and Vishny (1998) suggest that investors initially underreact to news due to conservatism. They then overreact over longer periods due to representativeness. In representativeness, as identified by Tversky and Kahneman (1974), you see things that look familiar and draw parallels between events that are not the same. In the case of investors, when they view recent price strength, they may assume it is the harbinger of favorable future economic conditions.